

Bivariate Data & multivariate Data:-

- ↓
- 2 variables & focuses on finding relationship b/w them.

Ex:- No of shirts & sales data

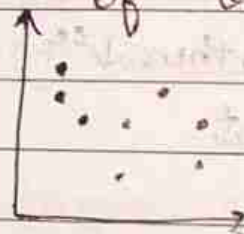
No of shirts	Sales of shirts
5	200
10	150
15	140

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- To analyze relationship, a scatter plot is used.

- It helps visualize trends; differences & relationships b/w 2 variables.
- Shows strength, shape, direction & presence of outliers

Ex:-



Scatter plot



Line graphs are similar to scatter plots.

Bivariate statistics:-

Covariance & correlation are ex's of Bivariate statistics

* Co-variance $\rightarrow \text{cov}(x, y)$

$\rightarrow$ used to measure the variance b/w 2D.

$$\text{Formula: } \text{cov}(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))(y_i - E(y))$$

where x_i, y_i = data values from x & y

$E(x)$ & $E(y) \rightarrow$ mean values of x_i & y_i

$N \rightarrow$ no. of given data

* Correlation $\rightarrow (r)$

Pearson correlation co-efficient (r) measures strength & direction of linear relationship betⁿ 2 variables.

$$\text{Formula: } r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \quad \sigma_x, \sigma_y \rightarrow \text{population std deviation}$$

$$\sigma_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

NOTE: If $r > 0$, x & y increase together.

If $r < 0$, one $\uparrow$ while the other $\downarrow$.

If $r = 0$, they are independent.

* Multivariate statistics:-

$\rightarrow$ Multivariate data involves more than two variables & is commonly used in ML.

$\rightarrow$ Similar to bivariate data but with multiple dependent variables.

Common analysis techniques:- Regression analysis, PCA, Path analysis.

Ex:-

id	Attribute1	Attribute2	Attribute3
1	1	4	1
2	2	5	2
3	3	6	3

$\rightarrow$ Above, there are 3 attributes, and their mean is called a centroid.

$\rightarrow$ Variance of multivariate data is represented as a covariance matrix.

* Heatmap:- Graphical representation of a 2D matrix using colors.

$\rightarrow$ Darker colors = larger values.

Lighter colors = smaller values.

$\rightarrow$ Helps in perceiving patterns easily.

Ex:- health data (weight vs health status)

* Pairplot:- A visualization tool for multivariate data.

$\rightarrow$ Consists of scatter plots for each pair of variables.

$\rightarrow$ Helps in finding relationships & correlations.

Ex:- A random matrix of 3 columns is plotted as a pairplot to show relationship.

* Linear Systems & Gaussian Elimination for multivariate Data:-

→ A Linear system of eq's → group of eq's with unknown variables.

• Written as $Ax=y$
solved as $x=A^{-1}y$, if A is non-zero

→ If a unique solutⁿ exists, the system is consistent independent.

→ If a multiple solutⁿ exists, the system is consistent dependent (No solutⁿ exists)

→ Gaussian Elimination solves large systems using row operations:

i. Convert the system into an augmented matrix
ii. Use a pivot element to eliminate variables in other rows.

iii. Convert to row echelon form.

iv. Use back-substitution to find unknowns.

Q. Given set of eqns, $2x_1 + 4x_2 = 6$
 $4x_1 + 3x_2 = 7$

Solve by Gaussian Elimination method.

Solu:- Let A = Augmented matrix

$$\Rightarrow A = \left[\begin{array}{cc|c} 2 & 4 & 6 \\ 4 & 3 & 7 \end{array} \right]$$

$$\Rightarrow A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 3 & 7 \end{array} \right] \quad R_1 \rightarrow R_1 / 2$$

$$R_2 = R_2 - 4R_1$$

$$\Rightarrow A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -5 & -5 \end{array} \right]$$

$$R_2 \rightarrow R_2 / -5$$

$$\Rightarrow A = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 1 \end{array} \right]$$

$$\Rightarrow R_1 \rightarrow R_1 - 2R_2$$

$$\Rightarrow A = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right]$$

$$\text{Now, } x_1 = 1 \quad x_2 = 1$$

* Matrix decomposition:-

Matrix decomposition breaks a matrix into smaller parts for easier calculations.

i. Eigen decomposition:-

Matrix A is decomposed into eigenvalues & eigen vectors:-

$$[A = Q \Lambda Q^T]$$

where $\rightarrow A \rightarrow$ diagonal matrix

$Q \rightarrow$ eigenvector matrix

$Q^T \rightarrow$ transpose of Q

ii. LU Decomposition:- splits a matrix into

$$A = LU$$

where $L \rightarrow$ lower triangular matrix
 $U \rightarrow$ Upper triangular matrix.

Steps:-

- i. Starts with an identity matrix.
- ii. Apply Gaussian elimination.
- iii. Reduce matrix to get L & U .

Q. Find LU decomposition of given matrix:-

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix}$$

$$\text{Soln:- } \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix}$$

$$R_2 = R_2 - 3R_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 3 & 4 & 2 \end{bmatrix}$$

$$R_3 = R_3 - 3R_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & -2 & -10 \end{bmatrix}$$

$$R_3 = R_3 - \frac{2}{3} R_2$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 2 \\ 3 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 2/3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & -10 \\ 0 & 0 & -10/3 \end{bmatrix}$$

$$\Rightarrow A = [L * U]$$

ML & Importance of Probability & Statistics:-

ML is closely linked to probability & statistics. Probability helps in understanding data distributions, while statistics assist in data analysis.

Types of Probability Distribution:-

1. Discrete Probability Distribution:- Deals with countable values.
2. Continuous Probability Distribution:- Deals with infinite possible values.

* Continuous Probability Distributions:-

- i. Normal Distribution (Gaussian distribution):-
 - Bell-shaped curve, most common distribution
 - mean, median & mode are equal.
 - used for heights, blood pressure & test scores.

Formula: $f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Z-score $\Rightarrow z = \frac{x-\mu}{\sigma}$

ii. Exponential Distribution:-

→ Used for modeling time until an event happens (Eg: waiting time for a bus)

→ Special case of the Gamma distribution

Formula $\rightarrow f(x, \lambda) = \lambda e^{-\lambda x}, x \geq 0$

Mean = $\frac{1}{\lambda}$ & Std. deviation = $\frac{1}{\lambda}$

* Discrete Probability Distribution:-

i. Binomial Distribution:-

Models the probability of 'x' successes in 'n' trials (Eg: coin flips)

→ Each trial has 2 outcomes: success (p) or failure (1-p) $\rightarrow q$

Formula: $P(x=x) = nC_x \cdot p^x \cdot q^{n-x}$ where $q = 1-p$

mean = np, variance = npq

iii. Poisson Distribution:- Models the no. of events in a fixed time (Eg: no of emails per hr)

→ Used when events are independent.

Formula: $P(x=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

Mean = λ , Std deviation = $\sqrt{\lambda}$

iii. Bernoulli Distribution:- Models an experiment with only 2 outcomes: success (1) or failure (0)

→ Used in binary classification problems.

Formula:

$f(k; p) = \begin{cases} 1-p, & k=0 \\ p, & k=1 \end{cases}$

mean = p, variance = pq

* Density Estimation:-

It is a process of estimating the probability density function $p(x)$ from these observations.

→ If $p(x)$ is small, x may be an anomaly.

→ There are 2 main types:-

i. Parametric Density Estimation

ii. Non-parametric Density Estimation

* k-Nearest Neighbours (KNN) Density Estimation:-

- Uses k nearest neighbours instead of fixed region size
- Density estimate is avg of values returned by k -nearest neighbours.

* Feature Engineering & Dimensionality Reduction Techniques:-

i. Feature Engg:- Selecting or transforming features to improve model performance.

- Feature selection:- choosing relevant features while removing unnecessary ones.
- Feature transformation:- Creating new features (Eg: BMI from height & weight)

ii. Feature subset selection:- Reduces dataset size by removing irrelevant & redundant features.

- Avoids the curse of dimensionality, where too many features increase complexity.
- Uses a greedy search approach to find the best subset.

iii. Feature removal criteria:-

- Relevancy:- Important features contribute more (Eg: mole vs nose in face detection)

- Redundancy:- Features providing duplicate info can be removed.

Selection process:-

1. Generate all possible subsets.
2. Evaluate subsets & model performance.
3. Select the best subset.

Selection methods:-

- * Filter-based:- uses statistical measures (correlation, mutual info)
- * Wrapper-based:- uses classifiers for feature selection, offering better performance but higher computation.

* Algorithms:- PCA reduces dimensions by transforming data into new features with max variance.

Steps:-

- i. Subtract mean from dataset.
- ii. Compute co-variance matrix.
- iii. Calculate eigenvalues & eigenvectors.
- iv. Select top eigenvectors for transformation.
- v. Transform dataset using selected eigen vectors.

- (2) Stepwise forward selection:- Starts with no attributes & adds the best one at each step until a good set is formed.

3. Stepwise Backward Elimination:-
Starts with all attributes & removes the worst one at each step until a good set is formed.

4. (LDA) Linear Discriminant Analysis:- LDA reduces dimensions while preserving class separability.

Steps:-

- Compute class means
- Compute b/w-class & within class scatter matrices
- Find eigenvectors & eigenvalues
- Transform data using top eigen vectors

5. Singular value Decomposition (SVD):- is a decomposition technique

$$A = USV^T$$

where U & V are orthogonal matrices

S is diagonal matrix with singular values

Steps:-

- Compute AA^T
- Find its eigenvalues & eigenvectors to form matrix U
- Sort eigenvalues in descending order
- Take the square root of eigenvalues to form diagonal matrix S
- Find eigenvalue & eigenvectors of $A^T A$ to get matrix V

☆ Design of a learning system:-

A learning system is built around a learning algo & contains these steps:-

1. Choosing a training experience:-

- In direct experience, correct move in a chess game are given
- In indirect experience, only move sequences & results are given
- The system learns better when training & testing samples have same distribution

2. Choosing a target function:-

- Defines what the system needs to learn
- In indirect experience, all legal moves are scored, & the scoring move is chosen

3. Representing a target function:-

- The knowledge can be stored in tables, rules or neural nets

- Simple linear function can be:

$$V = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

where x_1, x_2, x_3 are board features

w_0, w_1, w_2, w_3 are weights

4. Choosing an approximation algo:-

The goal is to adjust weights to minimize error

$$E = \sum_{\text{Training samples}} [V_{\text{train}}(b) - \hat{V}(b)]^2$$

* Introduction to concept learning:-

Concept learning is a way to learn patterns & generalizations from training data. It helps classify objects based on common features.

Concept learning process:-

- Input:- Training dataset with labeled examples
- Output:- A target function that identifies key features
- Test:- New instances are tested to see if they match the concept.

ex:- Identifying an elephant using features like trunk, fur, size & color.

Representation of a hypothesis:-

A hypothesis (h) is an approximation of a target function that maps input attributes to output values.

- Hypothesis Space: Includes all possible hypothesis for a target function.
- Version space: The set of hypothesis that are consistent with training data.
- Heuristic Space Search:- Heuristic search finds an optimized hypothesis using techniques like
 - Hill climbing
 - Genetic algos
 - Best-first search
 - A* algo
- Generalization:- This learning methodology will search through the hypothesis space for an approximate hypothesis by generalizing most specific hypothesis.
- Specialization:- This learning methodology will search through hypothesis space for an approximate hypothesis by specializing most general hypothesis.

Hypothesis space search by Find-S algm:-

Algm:-

Input:- Positive instances in the training dataset.

Output:- Hypothesis 'h'

Step: 1. Initialize 'h' to the most specific hypothesis
 $h = \langle \phi \ \phi \ \phi \ \phi \dots \rangle$

Step: 2. Generalize the initial hypothesis for the first positive instance.

Step: 3. For each subsequent instances:

If it is a positive instance

Check for each attribute value in the instance with hypothesis 'h'

If the attribute value is the same as the hypothesis value, then do nothing.

Else if attribute value is different than the hypothesis value, change it to '?' in 'h'

Else if it is a negative instance, ignore it.

Candidate Elimination:-

Algm:-

Input: Set of instances in the training dataset.

Output: Hypothesis G & S

1. Initialize G to the maximally general hypothesis.
2. Initialize S to the maximally specific hypothesis.
→ Generalize the initial hypothesis for first +ve instance.

3. For each subsequent new training instance,

i. If instance is +ve,

- Generalize S to include +ve instance

→ If the attribute value of +ve instance & S are different, fill that field value with '?'

→ If attribute value of +ve instance & S are same, then do not change.

- Prune G to exclude all inconsistent hypotheses in G with the +ve instance.

ii. If the instance is -ve,

- Specialize G to exclude -ve instances.

→ If the attribute value of S & the -ve instance are different, then fill that attribute value with S value.

→ If the attribute value of S & -ve instance are same, no need to update 'G' & fill that attribute with '?'

- Remove from S all inconsistent hypothesis with the -ve instance.

★ ML process:-

Performance:-

1. Choose an algm
2. Train the model on data
3. Tune parameters for better accuracy
4. Evaluate performance

Model Selection & Evaluation:-

Performance: How well the model fits training data.

Complexity: More complex models may overfit.

Re-sampling Methods [Improving Model Selection]:

1. Cross-validation: - Splitting data into subsets to train & validate models.
2. Hold out method: One-time split into training & test sets.
3. K-fold cross validation: - Data is split into k parts, training is $(k-1)$ times, & testing on remaining data.
4. Stratified K-fold cross validation: - ensures each fold maintains the same class

proportion.

5. Leave-one-out-cross validation (LOOCV):-
Each data point is tested once, using
rest for training.